ANSWER KEY

1. (*c*)	**2.** (*d*)	**3.** (*a*)	**4.** (*a*)	**5.** (*b*)	**6.** (*a*)	**7.** (*a*)	**8.** (*c*)	**9.** (*c*)	**10.** (*a*)
11. (*d*)	**12.** (*d*)	**13.** (*b*)	**14.** (*c*)	**15.** (*a*)	**16.** (*b*)	**17.** (*c*)	**18.** (*a*)	**19.** (*a,b,c*)	**20.** (*d*)
21. (*b*)	**22.** (*d*)	**23.** [45]	**24.** [9]	**25.** [4]	**26.** [2]	**27.** [4]	**28.** [324]	**29.** [25]	**30.** [3]

Hints and Solutions

1. (*c*) $\alpha,\beta=\dfrac{-\sqrt{6}\pm\sqrt{6-12}}{2}=\dfrac{-\sqrt{6}\pm\sqrt{6}i}{2}=\dfrac{\sqrt{6}}{2}(-1\pm i)$

$=\sqrt{3}\left(\dfrac{-1}{\sqrt{2}}\pm\dfrac{1}{\sqrt{2}}i\right)$

$=\sqrt{3}e^{\pm\frac{3\pi i}{4}}$

Required expression

$$=\frac{(\sqrt{3})^{23}\left(2\cos\frac{69\pi}{4}\right)+(\sqrt{3})^{14}\left(2\cos\frac{42\pi}{4}\right)}{(\sqrt{3})^{15}\left(2\cos\frac{45\pi}{4}\right)+(\sqrt{3})^{10}\left(2\cos\frac{30\pi}{4}\right)}$$

$=(\sqrt{3})^{8}=81$

2. (*d*) $x^2-\sqrt{2}x+2=0$

Roots of the given equation are

$x=\dfrac{\sqrt{2}\pm\sqrt{2-8}}{2}=\dfrac{\sqrt{2}\pm\sqrt{6}i}{2}$

$\therefore\alpha=\dfrac{\sqrt{2}+\sqrt{6}i}{2}=\sqrt{2}e^{\frac{i\pi}{3}}\ \&\ \beta=\sqrt{2}e^{\frac{-i\pi}{3}}$

$\alpha^{14}=2^7e^{\frac{i14\pi}{3}}=128\left[e^{\frac{i2\pi}{3}}\right]$

$\beta^{14}=\left(2^7e^{-\frac{i14\pi}{3}}\right)=128\left[e^{\frac{-i2\pi}{3}}\right]$

$\therefore\alpha^{14}+\beta^{14}=128(2)\cos\left(\dfrac{2\pi}{3}\right)=-128$

3. (*a*) For $x\le 3$ or $x\ge 5$

$x^2-8x+15-2x+7=0$

$x=5+\sqrt{3}$ or $x=5-\sqrt{3}$ (reject)

For $3<x<5$, $x^2-8x+15+2x-7=0$

$x=4$ or $x=2$ (reject)

Hence, sum of all the roots $=5+\sqrt{3}+4=9+\sqrt{3}$

4. (*a*) $f(-2)+f(3)=0$. One root of $f(x)=0$ is (-1)

Let's assume other root to be α.

$\therefore f(x)=a(x+1)(x-\alpha)$

Given that $f(-2)+f(3)=0$

$a(-2+1)(-2-\alpha)+a(3+1)(3-\alpha)=0$

$\Rightarrow(-1)(-2-\alpha)+4(3-\alpha)=0$

$\Rightarrow 2+\alpha+12-4\alpha=0$

$\Rightarrow 14-3\alpha=0\Rightarrow\alpha=\dfrac{14}{3}$

$\therefore$ Sum of roots $=\dfrac{14}{3}-1=\dfrac{11}{3}$

5. (*b*)

$$x^2-\left(5+3^{\sqrt{\log_3 5}}-5^{\sqrt{\log_5 3}}\right)x+3\left(3^{(\log_3 5)^{\frac{1}{3}}}-5^{(\log_5 3)^{\frac{2}{3}}}-1\right)=0$$

$3^{\sqrt{\log_3 5}}=3^{\sqrt{\log_3 5}\cdot\sqrt{\log_3 5}\cdot\sqrt{\log_5 3}}=3^{\log_3 5\cdot\sqrt{\log_5 3}}$

$=\left(3^{\log_3 5}\right)^{\sqrt{\log_5 3}}=5^{\sqrt{\log_5 3}}$

$3^{\sqrt[3]{\log_3 5}}=3^{\log_3 5\cdot\sqrt[3]{(\log_5 3)^2}}=\left(3^{\log_3 5}\right)^{(\log_5 3)^{2/3}}$

$=5^{(\log_5 3)^{2/3}}$

So, equation is $x^2-5x-3=0$ and roots are α & β

$\{\alpha+\beta=5;\ \alpha\beta=-3\}$

New roots are $\alpha+\dfrac{1}{\beta}\ \&\ \beta+\dfrac{1}{\alpha}$

i.e., $\dfrac{\alpha\beta+1}{\beta}\ \&\ \dfrac{\alpha\beta+1}{\alpha}$ i.e., $\dfrac{-2}{\beta}\ \&\ \dfrac{-2}{\alpha}$

Let $\dfrac{-2}{\alpha}=t\Rightarrow\alpha=\dfrac{-2}{t}$

As $\alpha^2-5\alpha-3=0$

$$\Rightarrow \left(\frac{-2}{t}\right)^2 - 5\left(\frac{-2}{t}\right) - 3 = 0$$

$$\Rightarrow \frac{4}{t^2} + \frac{10}{t} - 3 = 0$$

$\Rightarrow 4 + 10t - 3t^2 = 0$

$\Rightarrow 3t^2 - 10t - 4 = 0$

i.e., $3x^2 - 10x - 4 = 0$

6. (*a*) *Given,* $x^2 = 1 - 2i$

So $\alpha^2 = 1 - 2i = \beta^2$

$\Rightarrow \alpha^8 = \beta^8$

Now, $|\alpha^8 + \beta^8| = 2|\alpha^8|$

$= 2|\alpha^2|^4$

$= 2|\alpha^2|^4$

$= 2|\alpha^2|^4$

$= 2 \times 25 = 50$

7. (*a*)

8. (*c*) $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$

$$\Rightarrow e^{6x} - 2e^{3x} + 1 - e^x\left[e^{3x} + 12e^x - 1\right] = 0$$

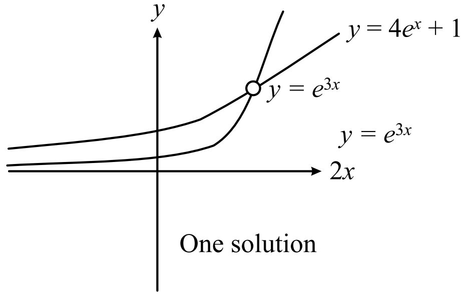
$$\Rightarrow \left(e^{3x} - 1\right)^2 - e^x\left(e^{3x} - 1\right) - 12e^{2x} = 0$$

$$\left(e^{3x} - 1\right)^2 - 4e^x\left(e^{3x} - 1\right) + 3e^x\left(e^{3x} - 1\right) - 12e^{2x} = 0$$

$$\left(e^{3x} - 1\right)\left\{e^{3x} - 1 - 4e^x\right\} + 3e^x\left\{e^{3x} - 1 - 4e^x\right\} = 0$$

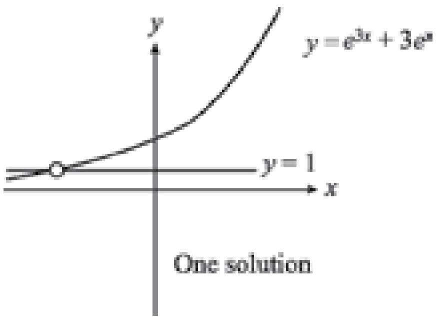
$$\left(e^{3x} - 4e^x - 1\right)\left(e^{3x} - 1 + 3e^x\right) = 0$$

Either $e^{3x} = 4e^x + 1$



One solution

OR $\left(e^{3x} = 1 - 3e^x\right)$



One solution

$\therefore$ The equation has total 2 solutions.

9. (*c*) Given, $x^2 + 3^{\frac{1}{4}}x + 3^{\frac{1}{2}} = 0$

Put $3^{1/4} = a$ *in the given equation, we get*

D : $x^2 + ax + a^2 = 0$

$$\Rightarrow \left(\frac{x}{a}\right)^2 + \frac{x}{a} + 1 = 0 \Rightarrow x = a\omega, a\omega^2$$

$\alpha = a\omega$, $\beta = a\omega^2$

$\alpha^{96} = a^{96}\omega^{96} = 3^{24}$

$\alpha^{12} - 1 = a^{12}\omega^{12} - 1 = 3^3 - 1 = 26$

$\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1) = 52 \times 3^{24}$

10. (*a*) Let $x = 4 + \cfrac{1}{5 + \cfrac{1}{4 + \cfrac{1}{5 + \cfrac{1}{4 + \cdots\infty}}}}$

Then,

$$x = 4 + \frac{1}{5 + \frac{1}{x}} \Rightarrow x - 4 = \frac{x}{5x + 1}$$

$\Rightarrow (x - 4)(5x + 1) = x$

$\Rightarrow 5x^2 - 20x - 4 = 0$

$$x = \frac{20 \pm \sqrt{400 + 4.5.4}}{2 \times 5} = \frac{20 \pm \sqrt{480}}{10}$$

But x is +*ve*.

$$\text{So, } x = \frac{20 + \sqrt{480}}{10} = \frac{20 + 4\sqrt{30}}{10} = 2 + \frac{2\sqrt{30}}{5}$$

11. (*d*) $\operatorname{cosec} 18° = \dfrac{1}{\sin 18°} = \dfrac{4}{\sqrt{5} - 1} = \sqrt{5} + 1$

Let $\operatorname{cosec} 18° = x = \sqrt{5} + 1$

$\Rightarrow x - 1 = \sqrt{5}$

Squaring both sides, we get

$x^2 - 2x + 1 = 5$

$\Rightarrow x^2 - 2x - 4 = 0$

12. (*d*) $a, b, c \rightarrow AP \Rightarrow 2b = a + c$ …(*i*)

Centroid is $\left(\frac{10}{3}, \frac{7}{3}\right)$

$$\frac{a + 2 + a}{3} = \frac{10}{3} \text{ and } \frac{c + b + b}{3} = \frac{7}{3}$$

$\Rightarrow 2a + 2 = 10$

$\Rightarrow 2a = 8$

$\Rightarrow a = 4;$

and $c + 2b = 7$ …(*ii*)

Put a = 4 in (*i*)

$2b = c + 4 \Rightarrow c - 2b + 4 = 0$ ...(*iii*)

Solving (*ii*) & (*iii*) we get

$$c = \frac{3}{2}$$

$\therefore\ 2b = c + 4 = \dfrac{11}{2}$

$\Rightarrow\ b = \dfrac{11}{4}$

$ax^2 + bx + 4 = 0 \Rightarrow 4x^2 + \dfrac{11x}{4} + 1 = 0$

$\alpha + \beta = \dfrac{-11}{4\times 4} = \dfrac{-11}{16};\ \alpha\beta = \dfrac{1}{4}$

$\therefore\ \alpha^2 + \beta^2 - \alpha\beta = (\alpha + \beta)^2 - 3\alpha\beta = \dfrac{121}{256} - \dfrac{3}{4} = \dfrac{-71}{256}$

13. (*b*) $\dfrac{\alpha^{3/8}}{\beta^{5/8}} + \dfrac{\beta^{3/8}}{\alpha^{5/8}} = \dfrac{\alpha+\beta}{(\alpha\beta)^{5/8}}$

For $x^2 - 64x + 256 = 0$

Sum of roots $= \alpha + \beta = 64$

Product of roots $= \alpha\beta = 256$

$\therefore\ \dfrac{\alpha+\beta}{(\alpha\beta)^{5/8}} = \dfrac{64}{\left(2^8\right)^{5/8}} = \dfrac{64}{32} = 2$

14. (c) α and β are roots of $5x^2 + 6x - 2 = 0$

$\Rightarrow 5\alpha^2 + 6\alpha - 2 = 0$

$\Rightarrow 5\alpha^{n+2} + 6\alpha^{n+1} - 2\alpha^n = 0$ ...(*i*)

(By multiplying α^n)

Similarly $5\beta^{n+2} + 6\beta^{n+1} - 2\beta^n = 0$ ...(*ii*)

By adding (i) and (ii)

$5S_{n+2} + 6S_{n+1} - 2S_n = 0$

For $n = 4$

$5S_6 + 6S_5 = 2S_4$

15. (*a*) Given α and β are the roots of the equation

$x^2 - x - 1 = 0$

$\Rightarrow\ \alpha + \beta = 1$ and $\alpha\beta = -1$

$\alpha^2 = \alpha + 1,\ \beta^2 = \beta + 1$

$\alpha^5 = 5\alpha + 3$

$\beta^2 = 5\beta + 3$

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$p_5 = 5(\alpha + \beta) + 6 = 5(1) + 6$

$p_5 = 11$ and $p_2 = \alpha2 + \beta2 = \alpha + 1 + \beta + 1$

$p_2 = 3$ and $p_3 = \alpha3 + \beta3 = 2\alpha + 1 + 2\beta + 1$

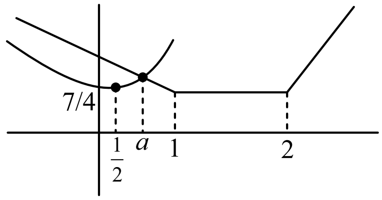
$= 2(1) + 2 = 4$

$p_2 \times p_3 = 12$ and $p_5 = 11 \Rightarrow p_5 \neq p_2 \times p_3$

16. (*b*) Let $3^x = t$

$t(t - 1) + 2 = |t - 1| + |t - 2|$

$t^2 - t + 2 = |t - 1| + |t - 2|$



are positive solution

$t = a$

$3^x = a$

$x = \log_3 a$ is singleton set.

17. (*c*) Using quadratic formula,

$$x = \frac{(\cos\theta\sin\theta + 1) \pm \sqrt{(\cos\theta\sin\theta + 1)^2 - 4\sin\theta\cos\theta}}{2\sin\theta}$$

$$= \frac{(\cos\theta\sin\theta + 1)^2 \pm (\cos\theta\sin\theta - 1)}{2\sin\theta}$$

$= \cos\theta$, cosec θ,

$\alpha = \cos\theta$, $\beta = \text{cosec}\ \theta$

$$\therefore\ \sum_{n=0}^{\infty} \alpha^n + \frac{(-1)^n}{\beta^n} = \sum_{n=0}^{\infty} (\cos\theta)^n + \sum_{n=0}^{\infty} (-\sin\theta)^n$$

$$= \frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}$$

18. (*a*) $x^2 + 2x + 2 = 0 \Rightarrow (x + 1)^2 = -1$

$$x = -1 \pm i = \sqrt{2}e^{i\left(\pm\frac{3\pi}{4}\right)}$$

$$\therefore \alpha^{15} + \beta^{15} = \left(\sqrt{2}\right)^{15} \times 2\cos\left(15 \times \frac{3\pi}{4}\right)$$

$$= 2^8\sqrt{2} \times \left(-\frac{1}{\sqrt{2}}\right) = -256$$

19. (*a*,*b*,*c*)

α, β are roots of $x^2 - x - 1$

$$a_{r+2} - a_r = \frac{(\alpha^{r+2} - \beta^{r+2}) - (\alpha^r - \beta^r)}{\alpha - \beta}$$

$$= \frac{(\alpha^{r+2} - \alpha^r) - (\beta^{r+2} - \beta^r)}{\alpha - \beta}$$

$$= \frac{\alpha^r(\alpha^2 - 1) - \beta^r(\beta^2 - 1)}{\alpha - \beta} = \frac{\alpha^r\alpha - \beta^r\beta}{\alpha - \beta}$$

$$= \frac{\alpha^{r+1} - \beta^{r+1}}{\alpha - \beta} = a_{r+1} \Rightarrow a_{r+2} - a_{r+1} = a_r$$

$$\Rightarrow \sum_{r=1}^{n} a_r = a_{n+2} - a_2 = a_{n+2} - \frac{\alpha^2 - \beta^2}{\alpha - \beta}$$

$a_{n+2} - (\alpha + \beta) = a_{n+2} - 1$

Now $\sum_{r=1}^{\infty}\frac{a_n}{10^n}=\frac{\sum_{r=1}^{\infty}\left(\frac{\alpha}{10}\right)^n-\sum_{r=1}^{\infty}\left(\frac{\beta}{10}\right)^n}{\alpha-\beta}$

$$\frac{\frac{\frac{\alpha}{10}}{1-\frac{\alpha}{10}}-\frac{\frac{\beta}{10}}{1-\frac{\beta}{10}}}{\alpha-\beta}=\frac{\frac{\alpha}{10-\alpha}-\frac{\beta}{10-\beta}}{(\alpha-\beta)}$$

$$=\frac{10}{(10-\alpha)(10-\beta)}=\frac{10}{89}$$

$$\sum_{n=1}^{\infty}\frac{b_n}{10^n}=\sum_{n=1}^{\infty}\frac{a_{n-1}+a_{n+1}}{10^n}=\frac{\frac{\alpha}{10}+}{1-\frac{\alpha}{10}}+\frac{\frac{\beta}{10}}{1-\frac{\beta}{10}}=\frac{12}{89}$$

Further, $b_n=a_{n-1}+a_{n+1}$

$$=\frac{(\alpha^{n-1}-\beta^{n-1})+(\alpha^{n+1}-\beta^{n+1})}{\alpha-\beta}$$

(as $\alpha\beta=-1\Rightarrow\alpha^{n-1}=-\alpha^n\beta$ & $\beta^{n-1}=-\alpha\beta^n$)

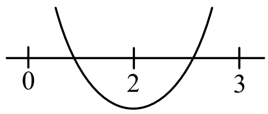
$$=\frac{\alpha^n(\alpha-\beta)+(\alpha-\beta)\beta^n}{\alpha-\beta}=\alpha^n+\beta^n$$

20. (*d*) One root lie in (0, 2)

So, $f(0)\,f(2)<0$

$(c-4)(c-24)<0$

$4(c-5)-4c+c-4<0$ $c\in(4, 24)$ …(*i*)



One root lie in (2, 3)

So, $f(2)\,f(3)<0$

$(c-24)(4c-49)<0$

$c\in\left(\frac{49}{4},24\right)$ …(*ii*)

Here (*i*) ∩ (*ii*)

$c\in\left(\frac{49}{4},24\right)$

21. (*b*) Let $(\sqrt{3}+\sqrt{2})^{x^2-4}=t$

$$t+\frac{1}{t}=10$$

$$\Rightarrow t=5+2\sqrt{6},5-2\sqrt{6}$$

$$\Rightarrow(\sqrt{3}+\sqrt{2})^{x^2-4}=5+2\sqrt{6},5-2\sqrt{6}$$

$$\Rightarrow x^2-4=2,-2\ \text{ or }\ x^2=6,2$$

$$\Rightarrow x=\pm\sqrt{2},\pm\sqrt{6}$$

22. (*d*) $x^2-4x+[x]+3=x[x]$

$\Rightarrow x^2-4x+3=x[x]-[x]$

$\Rightarrow(x-1)(x-3)=[x]\,.\,(x-1)$

$\Rightarrow x=1$ or $x-3=[x]$

$\Rightarrow x-[x]=3$

$\Rightarrow\{x\}=3$ (Not Possible) ({.} fractional part of x)

Only one solution $x=1$ in $(-\infty,\infty)$

23. [45] $\because$ **α, β be the roots of** $x^2+60^{\frac{1}{4}}x+a=0$

$\therefore\alpha+\beta=-60^{\frac{1}{4}}$ & $\alpha\beta=a$

Given $\alpha^4+\beta^4=-30\Rightarrow\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2=-30$

$$\Rightarrow\left\{(\alpha+\beta)^2-2\alpha\beta\right\}^2-2a^2=-30$$

$$\Rightarrow\left\{60^{\frac{1}{2}}-2a\right\}^2-2a^2=-30$$

$$\Rightarrow 60+4a^2-4a\times60^{\frac{1}{2}}-2a^2=-30$$

$$\Rightarrow 2a^2-4.60^{\frac{1}{2}}a+90=0$$

Product of roots $=\frac{90}{2}=45$

24. [9] $x^2-12x+[x]+31=0$

$\{x\}=x^2-11x+31$

$0\le x^2-11x+31<1$

$\Rightarrow x^2-6x-5x+30<0$

$\Rightarrow(x-6)(x-5)<0$

$\Rightarrow x\in(5, 6)$

So, $[x]=5$

$\Rightarrow x^2-12x+5+31=0$

$\Rightarrow x^2-12x+36=0$

$\Rightarrow(x-6)^2=0$

$\Rightarrow x=6$ but $x\in(5, 6)$

So, $x\in\phi$

Hence, $m=0$

$x^2-5|x+2|-4=0$

When $x\ge-2$

$x^2-5x-14=0$

$x^2-7x+2x-14=0$

$\Rightarrow(x-7)(x+2)=0\Rightarrow x=7,-2$

When $x<-2$

$x^2+5x+6=0$

$\Rightarrow(x+3)(x+2)=0$

$\Rightarrow x=-2,-3$

$\therefore x=\{7,-2,-3\}$

$\therefore n=3$

Now, $m^2+mn+n^2\Rightarrow n^2=9$

25. [4] Given, $p + q = 3$ and $p^4 + q^4 = 369$

$\therefore (p + q)^2 = 9$

$p^2 + q^2 = 9 - 2pq$

$$\frac{1}{\left(\frac{1}{p}+\frac{1}{q}\right)^2}=\frac{(qp)^2}{(q+p)^2}=\frac{(qp)^2}{9} \quad ...(i)$$

$\Rightarrow p^4 + q^4 = (p^2 + q^2)^2 - 2p^2q^2$

$\Rightarrow 369 = (9 - 2qp)^2 - 2(qp)^2$

$\Rightarrow 369 = 81 + 4p^2q^2 - 36pq - 2p^2q^2$

$\Rightarrow 288 = 2p^2q^2 - 36pq$

$\Rightarrow 144 = p^2q^2 - 18pq$

$\Rightarrow (pq)^2 - 2 \times 9 \times pq + 9^2 = 144 + 9^2$

$\Rightarrow (qp - 9)^2 = 225$

$\Rightarrow pq - 9 = \pm 15$

$\Rightarrow pq = \pm 15 + 9$

$pq = 24, -6$

(24 is rejected because $p^2 + q^2 = 9 - 2qp$ is negative)

From eq^n (i) we have $\frac{(pq)^2}{9} = 4$

26. [2] $e^{4x}+4e^{3x}-58e^{2x}+4e^{x}+1=0$

$e^{2x}+4e^{x}-58+4e^{-x}+e^{-2x}=0$

$$\left(e^{2x}+\frac{1}{e^{2x}}\right)+4\left(e^{x}+\frac{1}{e^{x}}\right)-58=0$$

$$\left(e^{2x}+\frac{1}{e^{2x}}+2\right)+4\left(e^{x}+\frac{1}{e^{x}}\right)=58+2$$

$$\left(e^{x}+\frac{1}{e^{x}}\right)^2+4\left(e^{x}+\frac{1}{e^{x}}\right)+4=64$$

$$\Rightarrow\left(e^{x}+\frac{1}{e^{x}}+2\right)^2=64$$

$\Rightarrow e^{x}+\frac{1}{e^{x}}+2=8$ (8 not possible)

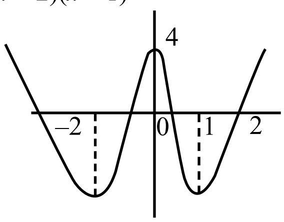
$e^{2x} - 6e^x + 1 = 0$

$$\therefore\ e^x=\frac{6\pm\sqrt{32}}{2}=3\pm2\sqrt{2}$$

27. [4] Let $f(x) = 3x^4 + 4x^3 - 12x^2 + 4$

So, $f'(x) = 12x^3 + 12x^2 - 24x = 12x(x^2 + x - 2)$

$= 12x(x + 2)(x - 1)$



So, number of distinct real roots = 4

28. [324] $P_n = \alpha^n + \beta^n$, $P_{n-1} = 11$ & $P_{n+1} = 29$

$P_n = \alpha^{n-2}.\alpha^2 + \beta^{n-2}.\beta^2 \quad ...(i)$

Now quadratic equation having roots α & β will be

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

i e $x^2 - x - 1 = 0$, put $x = \alpha$ and $= \beta$, ωε γετ,

$\alpha^2 = \alpha + 1$ and $\beta^2 = \beta + 1$

From equ (i), we have

$P_n = \alpha^{n-2}(\alpha + 1) + \beta^{n-2}(\beta + 1)$

$P_n = \alpha^{n-1} + \alpha^{n-2} + \beta^{n-1} + \beta^{n-2}$

$P_n = P_{n-1} + P_{n-2}$

So $P_{n+1} = P_n + P_{n-1}$

$\Rightarrow P_n = P_{n+1} - P_{n-1}$

$= 29 - 11 = 18$

$\Rightarrow P_n^2 = 324$

29. [25] Let $f(x) = (x - \alpha)(x - \beta)$

It is given that $f(0) = p \Rightarrow \alpha\beta = p$

and $f(1)=\frac{1}{3}\Rightarrow(1-\alpha)(1-\beta)=\frac{1}{3} \quad ...(i)$

Now, let us assume that α is the common root of $f(x) = 0$ and $fofofof(x) = 0$

$fofofof(x) = 0$

$\Rightarrow fofof(0) = 0 \quad [\because f(\alpha) = 0]$

$\Rightarrow fof(p) = 0 \quad [\because f(0) = p]$

So, $f(p)$ is either α or β.

$(p - \alpha)(p - \beta) = \alpha$

$(\alpha\beta - \alpha)(\alpha\beta - \beta) = \alpha \quad [\because P = \alpha\beta]$

$\Rightarrow (\beta - 1)(\alpha - 1)\beta = 1 \quad [\because$ Using eq^n (i)]

So, $\beta = 3$ ($\because a \neq 0$)

Now, $(1-\alpha)(1-3)=\frac{1}{3}$ [from eq^n (i)]

$$\Rightarrow \alpha=\frac{7}{6}$$

$$\text{So, } f(x)=\left(x-\frac{7}{6}\right)(x-3)$$

$$f(-3)=\left(-3-\frac{7}{6}\right)(-3-3)=25$$

30. [3] $x^8 - x^7 - x^6 + x^5 + 3x^4 - 4x^3 - 2x^2 + 4x - 1 = 0$

$\Rightarrow x^7(x - 1) - x^5(x - 1) + 3x^3(x - 1) - x(x^2 - 1) + 2x(1 - x) + (x - 1) = 0$

$\Rightarrow (x - 1)(x^7 - x^5 + 3x^3 - x(x + 1) - 2x + 1) = 0$

$\Rightarrow (x - 1)(x^7 - x^5 + 3x^3 - x^2 - 3x + 1) = 0$

$\Rightarrow (x - 1)(x^5(x^2 - 1) + 3x(x^2 - 1) - 1(x^2 - 1)) = 0$

$\Rightarrow (x - 1)(x^2 - 1)(x^5 + 3x - 1) = 0 \therefore x = \pm 1$ are roots of above equation and $x^5 + 3x - 1$ is a monotonic term hence vanishes at exactly one value of x other then 1 or – 1.

$\therefore$ 3 real roots.